

The Hardness of Proofs of Space Beyond Depth Robustness

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Section 1 fixes the pebbling model, states the two graph hypotheses and states the main theorem; section 2 proves the footprint and search lemmas it rests on; section 3 assembles them into the latency bound. Appendix A carries out the Filecoin-shaped specialization, and Appendix B maps every numbered item to the Lean 4 declaration that states or proves it.

1 Model and main result

1.1 Layered graphs and pebble configurations

Let G have layers $V_\ell, \dots, V_1$, each containing $n > 0$ nodes. We follow Reyzin's numbering [Rey24]: the top layer is V_1 , the bottom layer V_ℓ carries the targets, and the index j of a layer V_j is called its *level*. Edges are acyclic and are either intra-layer or go from V_{j-1} to V_j , so the analysis reads the stack from the challenge at level ℓ upward to level 1. For $X \subseteq V_j$, let $\text{pred}_j(X) \subseteq V_{j-1}$ be the set of inter-layer predecessors of X . An expansion profile $\beta : [0, 1] \rightarrow [0, 1]$ satisfies

$$|\text{pred}_j(X)| \geq \beta(|X|/n)n. \quad (1)$$

Fix $\pi \in (0, 1)$ and $\alpha_\pi > 0$. We assume uniform intra-layer depth robustness in the following direct form:

$$X \subseteq V_j, \quad |X| \geq \pi n \quad \implies \quad G[X] \text{ contains a directed path with at least } \alpha_\pi n \text{ nodes.} \quad (2)$$

The inequalities are over the reals, so they encode the required integer rounding.

A pebble configuration consists of black sets $\mathcal{B}_j \subseteq V_j$ and red sets $\mathcal{R}_j \subseteq V_j$. Put $\rho_j = |\mathcal{B}_j|/n$ and assume

$$\sum_{j=1}^{\ell} \rho_j \leq \rho, \quad |\mathcal{R}_j| \leq \delta n \quad \text{for every } j. \quad (3)$$

Black pebbles thus share one global budget; red pebbles have a per-layer budget. A node is *unpebbled* if it belongs to neither set in its layer.

For a challenge set $S \subseteq V_\ell$, let $F_j(S)$ be the nodes of V_j from which an unpebbled directed path reaches S , and write $w_j(S) = |F_j(S)|/n$. We require the challenge set to contain no red pebble and to satisfy

$$S \cap \mathcal{R}_\ell = \emptyset, \quad |S|/n \geq \zeta_\delta. \quad (4)$$

1.2 Expansion calculus

Define

$$\beta_\delta(x) := \beta(x) - \delta, \quad \text{gain}_\delta(x) := \beta(x) - \delta - x, \quad g_\pi := \text{gain}_\delta(\pi), \quad \bar{\pi} := 1 - \beta(\pi).$$

Let β be Reyzin's degree- d Chung expansion profile [Rey24, Claim 11]. It maps $[0, 1]$ into itself, satisfies $\beta(0) = 0$, is strictly increasing, and strictly expands every point of $(0, 1)$. It also satisfies the reversal identity [Rey24, Claim 13]

$$\beta(1 - \beta(x)) = 1 - x \quad (0 < x < 1). \quad (5)$$

We assume that β is concave and that $\beta(x) - x$ has a unique maximizer $\alpha_g < \pi$. Finally, $\alpha_\delta^{\min} \leq \alpha_g \leq \alpha_\delta^{\max}$ are the two specified zeros of gain_δ , and $g_\pi > 0$. A concave function on a segment is at least the smaller of its endpoint values [Rey24, Claim 15]; this makes gain_δ nonnegative on $[\alpha_\delta^{\min}, \alpha_\delta^{\max}]$, and it is the only way concavity is used below.

Choose a source weight $\sigma \in (\alpha_\delta^{\min}, \pi)$, and define the tracking quantities

$$\hat{\pi} := \min\{\bar{\pi}, \sigma, 1 - \beta(\sigma)\}, \quad \hat{g} := \min\{g_\pi, \text{gain}_\delta(\sigma)/2\}. \quad (6)$$

The reversal identity gives $\text{gain}_\delta(1 - \beta(\sigma)) = \text{gain}_\delta(\sigma)$. Hence concavity, applied between the two endpoints $\hat{\pi}$ and π , yields

$$\text{gain}_\delta(x) \geq \hat{g} \quad (\hat{\pi} \leq x \leq \pi). \quad (7)$$

Both $\hat{g}$ and $\beta_\delta(\pi) - \hat{\pi}$ are positive.

The source condition also says that the gain at σ is *twice* the tracking constant, and concavity carries that up an initial segment. Fix a *doubled-gain mid-point* $\tilde{\sigma}$ and the associated *growth potential* Φ :

$$\tilde{\sigma} \in [\sigma, \pi] \quad \text{with} \quad \text{gain}_\delta(\tilde{\sigma}) \geq 2\hat{g}, \quad \Phi(v) := \frac{\min\{v, \tilde{\sigma}\} - \sigma}{2\hat{g}} + \frac{\max\{v, \tilde{\sigma}\} - \tilde{\sigma}}{\hat{g}}. \quad (8)$$

By concavity applied between σ and $\tilde{\sigma}$, $\text{gain}_\delta \geq 2\hat{g}$ on all of $[\sigma, \tilde{\sigma}]$. Thus Φ measures progress from σ in levels, charging $2\hat{g}$ per level below $\tilde{\sigma}$ and $\hat{g}$ per level above it. The choice $\tilde{\sigma} = \sigma$ is always admissible and gives $\Phi(\pi) = (\pi - \sigma)/\hat{g}$, which is what the single-rate analysis charges; a genuine mid-point is what lemma 7 converts into a shorter growth phase. We use the scalar conditions

$$\alpha_\delta^{\min} < \zeta_\delta - \rho, \quad \zeta_\delta \leq \alpha_\delta^{\max}, \quad \rho > 0. \quad (9)$$

The first inequality keeps every challenge footprint in the positive-gain interval even after the entire black budget is spent. Together with concavity it also makes the gain at the challenge floor $\zeta_\delta - \rho$ positive, so

$$\tilde{g} := \min\{\text{gain}_\delta(\zeta_\delta - \rho), g_\pi\} > 0. \quad (10)$$

Write $\lceil x \rceil_+ := \max\{0, \lceil x \rceil\}$. Three capacities drive the theorem: the levels the searches can consume, the levels one extension attempt can span, and the number of chain breaks. The span has two bounds: h_0 assigns the whole budget ρ to each attempt, while h_1 uses disjointness to count that

budget once across the stack. Its growth term prices the growth phase by the better of the single-rate count and the potential of lemma 7. In order,

$$s := \left\lceil \frac{\pi - \zeta_\delta + \rho}{\widehat{g}} \right\rceil_+ + \left\lceil \frac{\rho}{\widehat{g}} \right\rceil - 1, \quad (11)$$

$$h_0 := \max \left\{ 1, \left\lfloor \frac{\pi - \sigma + \rho}{\widehat{g}} \right\rfloor \right\} + 2 \left\lceil \frac{\rho}{\widehat{g}} \right\rceil, \quad (12)$$

$$h_1 := 1 + \min \left\{ \max \left\{ 1, \frac{\pi - \sigma}{\widehat{g}} \right\}, \Phi(\pi) + 1 \right\}, \quad (13)$$

$$b^{\max} := \left\lceil \frac{\rho}{\beta_\delta(\pi) - \widehat{\pi}} \right\rceil - 1. \quad (14)$$

The corresponding level offsets are

$$s_0 := s + b^{\max} h_0, \quad s_1 := s + \frac{2\rho}{\widehat{g}} + b^{\max} h_1, \quad (15)$$

and the joint ledger of lemma 12 contributes a third,

$$s_2 := \begin{cases} \left(1 + \frac{\pi - \zeta_\delta}{\widehat{g}} \right)_+ + \frac{2\rho}{\min\{\widehat{g}, \widetilde{g}\}} & \text{if } b^{\max} = 0, \\ \infty & \text{otherwise,} \end{cases} \quad (16)$$

writing $(y)_+ := \max\{0, y\}$. These give the chain-length bound

$$z_{\min}(\ell) := \max \left\{ 1, \left\lceil \frac{\ell - s_1}{(b^{\max} + 1)h_1} \right\rceil_+, \left\lceil \frac{\ell - s_2}{h_1} \right\rceil_+, \left\lceil \frac{\ell - s_0}{(b^{\max} + 1)h_0} \right\rceil + 1 \right\}. \quad (17)$$

The third entry vanishes exactly when a break can be paid for, which is what $s_2 = \infty$ records: the joint ledger needs the searches and the attempts to occupy disjoint ranges of levels, and a restart search destroys that. All denominators are positive, so $s \geq 0$, $h_0 \geq 3$, $b^{\max} \geq 0$, $h_1 \geq 2$, $s_2 > 0$ and $s_0 \geq 0$. The four entries of (17) are independent lower bounds on the chain length, so taking their maximum loses nothing: s_1 and s_2 are not comparable in general, and neither dominates for every parameter choice. Every quantity in (10)–(17) is a function of $\beta, \delta, \pi, \rho, \sigma, \widetilde{\sigma}, \zeta_\delta$ alone; only $z_{\min}$ also reads ℓ . The three capacities are proved in lemmas 2 and 4 for s , in lemma 10 for h_0 , and in lemma 9 for $b^{\max}$; the joint constants in lemma 12; the global span h_1 is established inside lemma 11.

Theorem 1 (Latency theorem). *Assume (1)–(9) and $\sigma < \alpha_\pi$. If $\ell > s_0$, every pebble configuration satisfying (3) and every challenge set S satisfying (4) have an unpebbled directed path Q , contained in the footprint of S and ending in S , such that*

$$|Q| \geq \alpha_\pi n + (z_{\min}(\ell) - 1)(\alpha_\pi - \sigma)n. \quad (18)$$

In particular, for fixed parameters the path length is $\Omega(\ell n)$.

The proof first works with footprint bounds and then combines the resulting graph paths.

2 Footprint growth and search

2.1 Footprint bounds

The *footprint bound* started at level t with weight y is given by Reyzin's recurrence [Rey24]:

$$f_t = y, \quad f_{j-1} = \max\{0, \beta_\delta(f_j) - \rho_{j-1}\} \quad (j \leq t). \quad (19)$$

For the challenge we instead initialize at the bottom level,

$$f_\ell = \max\{0, \zeta_\delta - \rho_\ell\}. \quad (20)$$

Both run from their starting level upward, toward level 1.

2.2 The challenge floor and the first chain source

Call a challenge level j *fertile* when $f_j \geq \pi$. For $g > 0$, call a level t *g-expandable* when

$$\sum_{j=t-i}^{t-1} \rho_j \leq (i+1)g \quad \text{for every } i \geq 1. \quad (21)$$

This constrains the black weight in every window of levels lying above t .

Lemma 1 (General challenge floor). *Under (9), the challenge footprint bound satisfies, at every level m ,*

$$\zeta_\delta - \sum_{j=m}^{\ell} \rho_j \leq f_m \leq \alpha_\delta^{\max}. \quad (22)$$

Consequently $f_m \geq \zeta_\delta - \rho$, every gain $\text{gain}_\delta(f_m)$ is nonnegative, and every infertile level has gain at least the constant $\tilde{g} > 0$ of (10).

Proof. We prove both inequalities in (22) simultaneously [Rey24, Claims 4 and 5]; the same induction also gives the $\tilde{g}$ floor. They hold at $m = \ell$ by (20), the nonnegativity of ζ_δ , and $\zeta_\delta \leq \alpha_\delta^{\max}$. Suppose they hold at m . Since $\sum_{j=m}^{\ell} \rho_j \leq \rho$ by (3), the lower bound gives

$$f_m \geq \zeta_\delta - \rho > \alpha_\delta^{\min}.$$

Together with the upper bound, this places f_m in the active interval, so $\text{gain}_\delta(f_m) \geq 0$. The recurrence now advances the lower invariant.

For the upper invariant, β_δ is strictly increasing and fixes $\alpha_\delta^{\max}$, because $\text{gain}_\delta(\alpha_\delta^{\max}) = 0$. Hence $\beta_\delta(f_m) \leq \alpha_\delta^{\max}$; subtracting nonnegative black weight and truncating at zero preserves this upper bound. This completes the induction.

The floor $f_m \geq \zeta_\delta - \rho$ follows again from the total budget. If $f_m < \pi$, then $f_m \in [\zeta_\delta - \rho, \pi]$. Concavity of gain_δ implies that its value on this interval is at least the minimum of its endpoint values, namely $\tilde{g}$. $\square$

Lemma 2 (Infertile capacity). *The challenge footprint bound has at most*

$$\left\lceil \frac{\pi - \zeta_\delta + \rho}{\tilde{g}} \right\rceil_+$$

infertile levels in total. This is the first summand of s in (11).

Proof. Suppose there are $k > 0$ infertile levels, and let m be the smallest of them. The other $k - 1$ lie among the levels $m + 1, \dots, \ell$, each contributes gain at least $\tilde{g}$, and all remaining gains there are nonnegative by lemma 1. The accumulated bound [Rey24, Claim 3], applied at m with $f_\ell \geq \zeta_\delta - \rho_\ell$ and $f_m < \pi$, gives

$$\zeta_\delta - \sum_{j=m}^{\ell} \rho_j + (k - 1)\tilde{g} < \pi.$$

Since $\sum_{j=m}^{\ell} \rho_j \leq \rho$,

$$(k - 1)\tilde{g} < \pi - \zeta_\delta + \rho.$$

If $\zeta_\delta - \rho \geq \pi$, the floor from lemma 1 rules out an infertile level, and the displayed bound is 0. Otherwise, dividing by $\tilde{g} > 0$ and applying the ceiling gives the bound. The case $k = 0$ is immediate. $\square$

Lemma 2 replaces the black weight actually spent between the deepest infertile level and the challenge by the whole budget ρ . The joint ledger of lemma 12 needs the same count charged to the levels it is levied on, so we record that form as well.

Lemma 3 (Infertile capacity in a window). *Let $1 \leq p < \ell$. If $k \geq 1$ of the levels $p + 1, \dots, \ell$ are infertile for the challenge footprint bound, then*

$$(k - 1)\tilde{g} < \pi - \zeta_\delta + \sum_{j=p+1}^{\ell} \rho_j. \quad (23)$$

Consequently, for every $k \geq 0$,

$$k \leq \left(1 + \frac{\pi - \zeta_\delta}{\tilde{g}}\right)_+ + \frac{1}{\tilde{g}} \sum_{j=p+1}^{\ell} \rho_j, \quad (24)$$

whose first summand is the head of s_2 in (16).

Proof. This is the proof of lemma 2 with the last step omitted. Let $m \geq p + 1$ be the smallest infertile level among $p + 1, \dots, \ell$; the other $k - 1$ lie among $m + 1, \dots, \ell$, each contributes gain at least $\tilde{g}$, and all remaining gains there are nonnegative by lemma 1. The accumulated bound [Rey24, Claim 3] applied at m , with $f_\ell \geq \zeta_\delta - \rho_\ell$ and $f_m < \pi$, gives

$$\zeta_\delta - \sum_{j=m}^{\ell} \rho_j + (k - 1)\tilde{g} < \pi,$$

and $\sum_{j=m}^{\ell} \rho_j \leq \sum_{j=p+1}^{\ell} \rho_j$ because $m \geq p + 1$ and the black weights are nonnegative. This is (23). Dividing by $\tilde{g} > 0$ gives $k \leq 1 + (\pi - \zeta_\delta)/\tilde{g} + \sum_{j=p+1}^{\ell} \rho_j/\tilde{g}$, and the first two terms are at most their positive part. When $k = 0$, (24) holds because both terms on its right are nonnegative. $\square$

The same search runs in three places, on two different footprint bounds: at the start of the chain, after every break, and inside every extension attempt. Starting at a level t and going up, it

looks for a level that is both fertile and $\widehat{g}$ -expandable. At a level p it stops if p is such a level; if p is infertile it moves to $p - 1$; and if p is fertile but not $\widehat{g}$ -expandable, there is an $i \geq 1$ with

$$\sum_{j=p-i}^{p-1} \rho_j > (i+1)\widehat{g},$$

and it skips that *blocked* window, moving to $p - i - 1$. Blocked windows skipped this way occupy disjoint ranges of levels, so their black weight is charged once. The search may also stop because it has gone past the top of the graph, or past a prescribed cutoff. Nothing in the search uses what fertility means, so we may run it on any footprint bound. Running it on the challenge bound, and counting infertile levels with lemma 2, gives the first structural event.

Lemma 4 (First fertile expandable level). *If $\ell > s$, the search on the challenge footprint bound started at level ℓ stops at a level $b \geq \ell - s$ that is both fertile and $\widehat{g}$ -expandable.*

Proof. Let u count the levels the search skips as infertile and v those it skips inside blocked windows; at every point of its run it stands at level $\ell - u - v$. The infertile skips are at distinct infertile levels, so u is at most the count of lemma 2. If $v > 0$, summing the strict inequalities of the skipped windows over their disjoint ranges gives $v\widehat{g} < \rho$, whence $v \leq \lceil \rho/\widehat{g} \rceil - 1$. The two bounds add to s by (11), so the search never goes past level $\ell - s \geq 1$. It therefore stops inside the graph, at a fertile $\widehat{g}$ -expandable level $b \geq \ell - s$. $\square$

The challenge footprint at b has weight at least π . Condition (2) therefore gives an unpebbled path of length at least $\alpha_\pi n$ inside that footprint. Since $\sigma < \alpha_\pi$, take the first $\lceil \sigma n \rceil$ nodes of that path as a source set. Each source node begins a suffix that reaches S , and the source has normalized weight at least σ . This source set is the first *chain source*.

2.3 Tracking one source

Consider the footprint bound of a chain source, initialized by $f_t = \sigma$ at an expandable level t . The next lemma is the analytic reason for (6).

Lemma 5 (Mirror floor). *Let t be $\widehat{g}$ -expandable and let $m \leq t$. If the source footprint bound is at most π at every level of $m, \dots, t$, then it is at least $\widehat{\pi}$ at every level of $m, \dots, t$ — indeed at least σ .*

Proof. The claim at $m = t$ follows from $\widehat{\pi} \leq \sigma$. Proceed by strong induction downward. At the source, $\text{gain}_\delta(\sigma) \geq 2\widehat{g}$. At each of the $t - m - 1$ levels strictly between t and m , the induction hypothesis and the upper assumption place the bound in $[\widehat{\pi}, \pi]$, where (7) gives gain at least $\widehat{g}$. Thus the gain sum in the accumulated bound [Rey24, Claim 3] at m is at least $(t - m + 1)\widehat{g}$. Expandability with $i = t - m$ bounds $\sum_{j=m}^{t-1} \rho_j$ by the same quantity, and consequently $f_m \geq f_t = \sigma \geq \widehat{\pi}$. $\square$

Lemma 6 (Growth window). *Suppose $t_1 < t$ is the first level above t at which the source footprint bound is larger than π , and put $\rho_{\text{grow}} = \sum_{j=t_1+1}^{t-1} \rho_j$. Then*

$$t - t_1 \leq \max \left\{ 1, \left\lceil \frac{\pi - \sigma + \rho_{\text{grow}}}{\widehat{g}} \right\rceil \right\}. \quad (25)$$

Proof. If $t_1 = t - 1$, the claim follows from the explicit 1 in the maximum. Otherwise apply lemma 5 through level $t_1 + 1$. The source gain is at least $2\hat{g}$, and every later pre-crossing gain is at least $\hat{g}$. Since $f_{t_1+1} \leq \pi$, the accumulated bound [Rey24, Claim 3] gives

$$(t - t_1)\hat{g} \leq \pi - \sigma + \rho_{\text{grow}}.$$

The left side is an integer multiple of $\hat{g}$, so taking the floor proves the claim. $\square$

Lemma 6 charges every level of the growth phase at the single rate $\hat{g}$, which lower-bounds the gain throughout the tracking interval $[\hat{\pi}, \pi]$. In between the gain can be larger, and (8) certifies how much larger on an initial segment. A potential converts that into a level count.

Lemma 7 (Two-piece growth window). *In the situation of lemma 6,*

$$t - t_1 \leq \Phi(\pi) + 1 + \frac{\rho_{\text{grow}}}{\hat{g}}, \quad (26)$$

with Φ the growth potential of (8).

Proof. The potential is nondecreasing, $\Phi(\sigma) = 0$, and it is $1/\hat{g}$ -Lipschitz: its slope is $1/(2\hat{g})$ below $\tilde{\sigma}$ and $1/\hat{g}$ above, and $1/\hat{g}$ is the larger of the two.

One free level advances Φ by at least 1. Let $\sigma \leq v \leq \pi$. If $v \leq \tilde{\sigma}$ then $\text{gain}_\delta(v) \geq 2\hat{g}$ by (8) and concavity, so $\beta_\delta(v) - v \geq 2\hat{g}$; the increment $\Phi(\beta_\delta(v)) - \Phi(v)$ is at least $(\beta_\delta(v) - v)/(2\hat{g}) \geq 1$, because the slope of Φ on $[v, \beta_\delta(v)]$ is at least $1/(2\hat{g})$ throughout, whether or not that interval crosses $\tilde{\sigma}$. If $v \geq \tilde{\sigma}$ then (7) gives $\beta_\delta(v) - v \geq \hat{g}$, and the slope of Φ is exactly $1/\hat{g}$ there, so the increment is again at least 1.

Telescoping. Put $m_i = t - i$ and $N = t - t_1 - 1$, so that $f_{m_i} \leq \pi$ for $0 \leq i \leq N$. By lemma 5 each such f_{m_i} is at least σ , so the previous paragraph applies at every step. The recurrence (19) gives $f_{m_{i+1}} \geq \beta_\delta(f_{m_i}) - \rho_{m_{i+1}}$, and monotonicity together with the Lipschitz bound yields

$$\Phi(f_{m_{i+1}}) \geq \Phi(\beta_\delta(f_{m_i})) - \frac{\rho_{m_{i+1}}}{\hat{g}} \geq \Phi(f_{m_i}) + 1 - \frac{\rho_{m_{i+1}}}{\hat{g}}.$$

Summing from $i = 0$ to $N - 1$, and using $\Phi(f_t) = \Phi(\sigma) = 0$,

$$\Phi(f_{t_1+1}) \geq N - \frac{1}{\hat{g}} \sum_{j=t_1+1}^{t-1} \rho_j = N - \frac{\rho_{\text{grow}}}{\hat{g}}.$$

Since $f_{t_1+1} \leq \pi$ and Φ is nondecreasing, the left side is at most $\Phi(\pi)$, so $t - t_1 = N + 1 \leq \Phi(\pi) + 1 + \rho_{\text{grow}}/\hat{g}$. $\square$

Comparing the two window bounds is a one-line computation:

$$\Phi(\pi) + \frac{\tilde{\sigma} - \sigma}{2\hat{g}} = \frac{\pi - \sigma}{\hat{g}}, \quad (27)$$

so the potential saves $(\tilde{\sigma} - \sigma)/(2\hat{g})$ levels and pays back the one free level that lemma 5 extracts at the source. It therefore wins exactly when $\tilde{\sigma} - \sigma > 2\hat{g}$, and the growth term $h_1 - 1$ of (13) takes whichever bound is smaller. Both lemmas 6 and 7 hold, so

$$t - t_1 \leq h_1 - 1 + \frac{\rho_{\text{grow}}}{\hat{g}} \quad (28)$$

in either case; this is the only form the accounting uses.

Once a footprint bound is fertile, the same search has a local bound depending only on the black weight inside its window.

Lemma 8 (Fertile continuation). *Let $f_{t_1} \geq \pi$, and suppose that down to a cutoff e the bound stays in $[\hat{\pi}, \alpha_\delta^{\max}]$. The search beginning at t_1 ends at a level t_2 , either by leaving $(e, t_1]$ or by finding a fertile $\hat{g}$ -expandable level, and*

$$t_1 - t_2 \leq \max \left\{ 0, 2 \left\lceil \frac{\rho_{\text{cont}}}{\hat{g}} \right\rceil - 1 \right\}, \quad \rho_{\text{cont}} := \sum_{j=t_2+1}^{t_1-1} \rho_j. \quad (29)$$

Proof. Run the same search, now calling a level fertile when the tracked bound is at least π there, and stopping also when it reaches e . Let u and v denote its infertile and blocked charges. All gains in the search are nonnegative. At an infertile level the value lies in $[\hat{\pi}, \pi]$, so its gain is at least $\hat{g}$. If $u > 0$, apply the accumulated bound [Rey24, Claim 3] at the last infertile level the search meets. The initial term $f_{t_1} - \pi$ is nonnegative, so $(u - 1)\hat{g} < \rho_{\text{cont}}$, giving $u \leq \lceil \rho_{\text{cont}}/\hat{g} \rceil$. The disjoint blocked ranges satisfy $v\hat{g} < \rho_{\text{cont}}$ when $v > 0$. If $\rho_{\text{cont}} = 0$, these strict inequalities force $u = v = 0$. Otherwise $v \leq \lceil \rho_{\text{cont}}/\hat{g} \rceil - 1$. Since the search goes up exactly through the levels counted by $u + v$, in the positive case

$$t_1 - t_2 \leq u + v \leq 2 \left\lceil \frac{\rho_{\text{cont}}}{\hat{g}} \right\rceil - 1.$$

This is the stated bound in both cases. $\square$

The cutoff in lemma 8 is the first level at which the tracked bound drops below $\hat{\pi}$, or the top of the graph if that never happens. The former event has a fixed price.

Lemma 9 (Break charge). *Suppose $f_{t_1} \geq \pi$, $m < t_1$, all intermediate gains are nonnegative, and $f_m < \hat{\pi}$. Then*

$$\beta_\delta(\pi) - \hat{\pi} < \sum_{j=m}^{t_1-1} \rho_j. \quad (30)$$

Proof. Strict monotonicity of β_δ and the first recurrence step give

$$f_{t_1-1} \geq \beta_\delta(f_{t_1}) - \rho_{t_1-1} \geq \beta_\delta(\pi) - \rho_{t_1-1}.$$

Apply the accumulated bound [Rey24, Claim 3] from $t_1 - 1$ down to m . Discarding the nonnegative intermediate gains yields

$$f_m \geq \beta_\delta(\pi) - \sum_{j=m}^{t_1-1} \rho_j.$$

Combining this with $f_m < \hat{\pi}$ proves (30). $\square$

The two phases together form one extension attempt.

Lemma 10 (One extension attempt). *An extension attempt from any $\hat{g}$ -expandable chain source at level t ends at the top of the graph, at the next chain source, or at a break, and its terminal level m satisfies*

$$t - m \leq h_0 - 1, \quad (31)$$

with h_0 the integer (12). In the break outcome, its window also satisfies (30).

Proof. Follow the source footprint bound until it first exceeds π , or until the graph is exhausted. The proof of lemma 6, with the terminal level in place of t_1 in the latter case, bounds the growth phase by $\max\{1, \lfloor (\pi - \sigma + \rho)/\widehat{g} \rfloor\}$, and (28) bounds it by $h_1 - 1 + \rho_{\text{grow}}/\widehat{g}$. If a fertile level is reached, run the search of lemma 8 until it finds a fertile expandable level, reaches the top, or crosses the tracking floor. These are exactly the three stated outcomes. The continuation phase has length at most $2\lceil \rho/\widehat{g} \rceil - 1$ by lemma 8 and the global budget. Adding the two phase bounds gives (31). (If the graph ends during the growth phase, the continuation contribution is zero.) If the tracking floor is crossed, lemma 9 supplies the additional charge. Finally, at a fertile expandable outcome, the corresponding source footprint has weight at least π , so (2) constructs the next chain source exactly as after lemma 4. $\square$

3 The latency theorem

3.1 Building and counting chains

A *chain source* is a source set of weight at least σ at an expandable level such that every node in the set has an unpebbled path to S . A *chain segment* begins at a fertile expandable level found by searching the challenge footprint bound; an unpebbled path in the corresponding footprint supplies the first chain source. Starting from the last chain source, repeatedly apply lemma 10. A successful attempt finds the next chain source of the same segment. At a break, abandon that segment and resume the search on the challenge footprint bound from the break level. All searches and attempts go up the stack, so their charged intervals are pairwise disjoint.

Let z be the largest number of chain sources in any completed segment, a the number of extension attempts, and b the number of breaks.

Lemma 11 (Counting sources and levels). *Run the construction from level ℓ and suppose that its initial search finds the first chain source. It then terminates with $z \geq 1$ and natural numbers a, b satisfying*

$$1 + a \leq (b + 1)(z + 1), \quad (32)$$

$$\ell \leq s + a(h_0 - 1), \quad (33)$$

$$\ell \leq s + \frac{2\rho}{\widehat{g}} + ah_1, \quad (34)$$

$$b \leq b^{\max}. \quad (35)$$

One of the resulting segments contains z chain sources, each obtained from unpebbled paths in the corresponding footprints. If $b = 0$, that segment is the only one and its last chain source lies at a level m with

$$m \leq h_0 - 1. \quad (36)$$

Proof. Every level of the graph is charged either to a search or to an extension attempt. Write u for the number of one-step advances past an infertile level and v for the number of levels charged to blocked search windows, so the searches consume $u + v$ levels in total. An extension attempt increases a by one and, by lemma 10, advances at most $h_0 - 1$ levels. Hence $\ell \leq u + v + a(h_0 - 1)$.

Now $u + v \leq s$. Restart searches operate on disjoint level ranges, so their infertile moves are distinct infertile levels of the single challenge footprint bound, and lemma 2 bounds u by the first

summand of (11). The blocked windows are disjoint as well; if their total length is $v > 0$, summing their defining strict inequalities gives

$$v\hat{g} < \sum_{j \text{ in blocked windows}} \rho_j \leq \rho \leq \left\lceil \frac{\rho}{\hat{g}} \right\rceil \hat{g},$$

so $v \leq \lceil \rho/\hat{g} \rceil - 1$, which holds trivially when $v = 0$. Adding the two bounds gives (33).

For the global version, do not replace the spend of each attempt by the whole budget. The growth phase of an attempt that sees black weight ρ_{grow} spans at most $h_1 - 1 + \rho_{\text{grow}}/\hat{g}$ levels, and its continuation phase, with weight ρ_{cont} , spans at most $1 + 2\rho_{\text{cont}}/\hat{g}$. The attempt windows are disjoint, so summing gives

$$\ell \leq u + v + ah_1 + \frac{1}{\hat{g}} \sum \rho_{\text{grow}} + \frac{2}{\hat{g}} \sum \rho_{\text{cont}} \leq s + ah_1 + \frac{2\rho}{\hat{g}},$$

which is (34).

Initially there is one chain source. A successful attempt adds one chain source to the current segment. A terminal or broken attempt contributes no chain source, and a break increases the number of segments by one. If the numbers of chain sources in the segments are $z_1, \dots, z_{b+1}$, then

$$1 + a \leq \sum_{j=1}^{b+1} (z_j + 1) \leq (b+1)(z+1),$$

which proves (32). The same induction retains the footprints and unpebbled paths used to obtain each new chain source, including every chain source of a segment attaining z .

For (35), every break window spends strictly more than $\beta_\delta(\pi) - \hat{\pi}$ by lemma 9, and break windows are disjoint. If $b > 0$, summing over them gives $b(\beta_\delta(\pi) - \hat{\pi}) < \rho \leq (b^{\max} + 1)(\beta_\delta(\pi) - \hat{\pi})$, so $b \leq b^{\max}$; this holds trivially when $b = 0$.

Finally, every nonterminal attempt either finds the next chain source or causes a restart, and a search that leaves the graph is terminal. Since the level strictly decreases, the construction terminates after finitely many events.

For (36), suppose $b = 0$. There is then one segment, and the construction can only stop by leaving the graph, which an attempt does. That attempt starts at the last chain source at level m and ends at a terminal level $m' \leq 0$; by (31), $m - m' \leq h_0 - 1$, and $m' \leq 0$ gives $m \leq h_0 - 1$. $\square$

Lemma 11 charges the black budget ρ three separate times. It appears in the infertile-capacity term of s through lemma 2, again in the blocked-window term $\lceil \rho/\hat{g} \rceil - 1$ of s , and twice more in the $2\rho/\hat{g}$ of (34). Yet the levels those three charges are levied on are pairwise disjoint — a level belongs either to a search or to an attempt, and the construction moves strictly upward — so one interval sum bounds all three, and (3) caps that sum by ρ . Collecting them is the joint ledger.

The obstacle in general is a break: the restart search that follows it runs above levels already charged to attempts, and the disjointness fails. When $b^{\max} = 0$ no break can be paid for, there is exactly one search, and it precedes every attempt.

Lemma 12 (Joint ledger). *Suppose $b^{\max} = 0$ and $\ell > s$. Then the construction terminates with a single segment, and its numbers $z \geq 1$ and a satisfy (32) and*

$$\ell \leq s_2 + ah_1. \tag{37}$$

Proof. By lemma 9 a break costs more than $\beta_\delta(\pi) - \hat{\pi}$ of black weight, and $b^{\max} = 0$ says ρ is at most that, so no attempt breaks. Hence there is one segment, and the only search is the initial one; it starts at level ℓ and, by lemma 4, stops at the first chain source $b_1 \geq \ell - s$. Every attempt window lies in $[1, b_1 - 1]$: the first attempt starts at b_1 and charges only the levels strictly above it, and every later attempt starts higher still, while the search occupied $b_1 + 1, \dots, \ell$. Write

$$\rho_{\text{search}} := \sum_{j=b_1+1}^{\ell} \rho_j, \quad \rho_{\text{attempts}} := \sum_{j=1}^{b_1-1} \rho_j,$$

so that $\rho_{\text{search}} + \rho_{\text{attempts}} \leq \rho$ by (3).

The search. Let u and v be as in the proof of lemma 4, so the search consumes $u + v$ levels and $\ell - b_1 = u + v$. Its infertile skips are at distinct infertile levels of $b_1 + 1, \dots, \ell$, so (24) with $p = b_1$ bounds u by the head of s_2 plus $\rho_{\text{search}}/\tilde{g}$. Its blocked windows are disjoint subranges of the same levels, so $v\hat{g} < \rho_{\text{search}}$ when $v > 0$, and $v \leq \rho_{\text{search}}/\hat{g}$ in either case. Writing ε for the head of s_2 in (16) and bounding both rates below by $\min\{\hat{g}, \tilde{g}\}$,

$$\ell - b_1 \leq \varepsilon + \frac{\rho_{\text{search}}}{\tilde{g}} + \frac{\rho_{\text{search}}}{\hat{g}} \leq \varepsilon + \frac{2\rho_{\text{search}}}{\min\{\hat{g}, \tilde{g}\}}. \quad (38)$$

The attempts. As in the proof of (34), an attempt whose growth phase sees black weight ρ_{grow} and whose continuation sees ρ_{cont} spans at most $h_1 - 1 + \rho_{\text{grow}}/\hat{g}$ levels growing, by (28), and at most $1 + 2\rho_{\text{cont}}/\hat{g}$ continuing, by lemma 8; that is $h_1 + \rho_{\text{grow}}/\hat{g} + 2\rho_{\text{cont}}/\hat{g}$ in total. The attempt windows are disjoint, so summing over the a attempts,

$$b_1 \leq a h_1 + \frac{2\rho_{\text{attempts}}}{\min\{\hat{g}, \tilde{g}\}}. \quad (39)$$

Adding (38) and (39) and using $\rho_{\text{search}} + \rho_{\text{attempts}} \leq \rho$ gives $\ell \leq \varepsilon + a h_1 + 2\rho/\min\{\hat{g}, \tilde{g}\}$, which is (37) by (16). The segment structure (32) is unchanged, and $z \geq 1$ because lemma 4 produced a chain source. $\square$

The four inequalities now give the chain length.

Lemma 13 (Chain length). *Whenever $\ell > s_0$, the construction finds a segment with*

$$z \geq z_{\min}(\ell) \quad (40)$$

chain sources.

Proof. If $\ell > s_0$, then in particular $\ell > s$, because $s_0 = s + b^{\max}h_0 \geq s$. Lemma 4 therefore finds the first chain source before the graph ends, and we may apply lemma 11. From (32) and (35),

$$a \leq (b^{\max} + 1)(z + 1) - 1 = (b^{\max} + 1)z + b^{\max}.$$

Substituting this into (33), and using $z \geq 1$ to make the last step strict, gives

$$\begin{aligned} \ell &\leq s + ((b^{\max} + 1)z + b^{\max})(h_0 - 1) \\ &< s + ((b^{\max} + 1)z + b^{\max})h_0 \\ &= s + b^{\max}h_0 + z(b^{\max} + 1)h_0 = s_0 + z(b^{\max} + 1)h_0. \end{aligned}$$

Thus the constant-charge entry of (17) is at most z .

Substitution in (34) instead gives

$$\ell \leq s + \frac{2\rho}{g} + ((b^{\max} + 1)z + b^{\max})h_1 = s_1 + z(b^{\max} + 1)h_1,$$

so the nonnegative-ceiling entry of (17) is also at most z .

For the third entry of (17) there is nothing to prove unless $b^{\max} = 0$, since $s_2 = \infty$ otherwise. When $b^{\max} = 0$, (32) reads $1 + a \leq z + 1$ and lemma 12 gives $\ell \leq s_2 + a h_1 \leq s_2 + z h_1$, so that entry is at most z as well. Finally $z \geq 1$, proving (40). $\square$

3.2 Combining the paths

Lemma 14 (Path length from a chain segment). *A segment of $z \geq 1$ chain sources yields an unpebbled path in the footprint of S of length at least*

$$p(z) := \alpha_\pi n + (z - 1)(\alpha_\pi - \sigma)n. \quad (41)$$

Proof. For the first chain source, depth robustness gives a path Q of length at least $\alpha_\pi n$ in the challenge footprint. The last node of Q has an unpebbled path to S ; appending it cannot shorten Q . This proves (41) for $z = 1$.

We maintain the stronger invariant that every node in the z th chain source has an unpebbled path to S of length at least $z(\alpha_\pi - \sigma)n$. It holds for the first chain source: if the source is the first $k = \lceil \sigma n \rceil$ nodes of Q , then a source position $i < k$ satisfies $i < \sigma n$, and its suffix contains at least

$$|Q| - i \geq \alpha_\pi n - \sigma n = (\alpha_\pi - \sigma)n$$

nodes.

Suppose the invariant holds for the z th chain source T , and a later fertile footprint $F_j(T)$ is used to obtain the next chain source. Let Q be the path given by depth robustness. Its last node reaches some $y \in T$ by an unpebbled connector R . The levels of the two endpoints differ, so $|R| \geq 2$. Append Q , R , and the old unpebbled path from y to S . Concatenation identifies two pairs of endpoints, while R supplies at least two nodes; hence the result has length at least

$$\alpha_\pi n + z(\alpha_\pi - \sigma)n = p(z + 1).$$

For a node in the new source prefix, use its suffix of Q instead of all of Q . That suffix contributes at least $(\alpha_\pi - \sigma)n$, and the old path contributes $z(\alpha_\pi - \sigma)n$. This proves the stronger invariant for the next chain source. All pieces lie in footprints, hence are unpebbled; acyclicity of the layered graph makes the concatenation a directed path rather than a walk with repeated nodes. $\square$

Proof of theorem 1. Let the constants be as in (11)–(17). Footprint bounds lower-bound the corresponding footprints, so every fertile challenge footprint has weight at least π . Thus (2) constructs the first chain source of each segment. The same applies to each source footprint, so every successful extension constructs the next chain source.

When $\ell > s_0$, lemma 13 gives a segment with $z \geq z_{\min}(\ell)$. By lemma 14, that segment yields an unpebbled path ending in S . Since $\sigma < \alpha_\pi$, the lower bound is increasing in z , and hence

$$|Q| \geq \alpha_\pi n + (z_{\min}(\ell) - 1)(\alpha_\pi - \sigma)n.$$

Every piece used in the concatenation belongs to the footprint of S . Since s_0, s_1, h_0 , and h_1 depend only on the scalar parameters, $z_{\min}(\ell) = \Omega(\ell)$, and the final claim follows. $\square$

References

- [Fis19] Ben Fisch. Tight proofs of space and replication. In Yuval Ishai and Vincent Rijmen, editors, *EUROCRYPT 2019, Part II*, volume 11477 of *LNCS*, pages 324–348. Springer, Cham, May 2019.
- [Rey24] Leonid Reyzin. Proofs of space with maximal hardness. In *65th FOCS*, pages 1159–1177. IEEE Computer Society Press, October 2024.

A Filecoin parameters

Let β_8 denote the degree-eight Chung expansion profile of Reyzin [Rey24, Claims 11 and 13]. We assume its concavity and unique-maximizer condition. The scalar parameters are the Filecoin-shaped SDR choice of Fisch [Fis19]; a subscript F marks this specialization.

The chosen parameters are

$$\delta_F = 0.0378, \quad \pi_F = 0.8, \quad \rho_F = 0.8, \quad \zeta_{\delta,F} = 0.8622, \quad \sigma_F = 0.1184, \quad \alpha_{\pi,F} = 0.2, \quad \tilde{\sigma}_F = 3/5. \quad (42)$$

The curve contributes $\beta_8(\pi_F) = 0.94911\dots$ and $\beta_8(\sigma_F) = 0.37888\dots$, so the two zeros of gain_δ are $\alpha_\delta^{\min} = 0.00976\dots$ and $\alpha_\delta^{\max} = 0.95243\dots$, and the expansion calculus of section 1 takes the values of table 1. Every integer there is certified in the formal development from the rational bracket

$$0.1113 < g_{\pi,F} < 0.1114 \quad (43)$$

alone; no floating-point evaluation of β_8 enters a proof.

Beyond (9), the specialization verifies

$$\bar{\pi} < \zeta_\delta - \rho, \quad \zeta_\delta \leq \alpha_\delta^{\max}, \quad \text{gain}_\delta(\sigma) \geq 2g_\pi, \quad \rho < \beta_\delta(\pi) - \bar{\pi}, \quad \text{gain}_\delta(\tilde{\sigma}) \geq 2g_\pi, \quad (44)$$

the last of which is (8) at $\hat{g} = g_\pi$, with $\text{gain}_\delta(3/5) = 0.23406\dots$ against $2g_{\pi,F} = 0.22262\dots$. Three of these entries carry the whole specialization. The condition $\text{gain}_\delta(\sigma) \geq 2g_\pi$ holds with little room, $\text{gain}_\delta(\sigma_F) = 0.22268\dots$ against $2g_{\pi,F} = 0.22262\dots$, and it collapses $\hat{g}_F$ to $g_{\pi,F}$. The room is small because this condition is what fixes σ_F from below: the per-link payoff $(\alpha_\pi - \sigma)n$ rewards a small σ , while gain_δ is still increasing there, so the least admissible source weight is $\sigma_{\min} = 0.11834\dots$ and the four-digit value 0.118 is already inadmissible. The challenge floor $\zeta_{\delta,F} - \rho_F = 0.0622$ sits above $\bar{\pi}_F$ and has gain $0.13476\dots > g_{\pi,F}$, which collapses $\tilde{g}_F$ as well. Finally, $\beta_\delta(\pi_F) - \hat{\pi}_F = 0.86042\dots$ exceeds the entire budget $\rho_F = 0.8$, so $b_F^{\max} = 0$ and no break can be paid for.

Corollary 1 (Filecoin-shaped specialization). *Assume inter-layer expansion according to β_8 and (2) with $\alpha_{\pi,F} = 0.2$. Whenever $\ell > 14$, every pebble configuration satisfying (3) and every challenge set satisfying (4) have a footprint containing an unpebbled path of length at least*

$$0.2n + (z_{\min,F}(\ell) - 1)0.0816n,$$

where, writing $h_{1,F} = 0.4408/g_{\pi,F} + 2$,

$$z_{\min,F}(\ell) = \max \left\{ 1, \left\lceil \frac{\ell - 14 - 1.6/g_{\pi,F}}{h_{1,F}} \right\rceil_+, \left\lceil \frac{\ell - 1 + 0.0622/g_{\pi,F} - 1.6/g_{\pi,F}}{h_{1,F}} \right\rceil_+, \left\lfloor \frac{\ell - 14}{29} \right\rfloor + 1 \right\}. \quad (45)$$

Table 1: The tracking constants and the capacities at the parameters (42). The middle column names the definition being evaluated.

Quantity	Definition	Value
$\bar{\pi}_F$	$1 - \beta_8(\pi)$	$0.05088 \dots$
$g_{\pi,F}$	$\text{gain}_\delta(\pi)$	$0.11131 \dots$
$\text{gain}_\delta(\sigma_F)$	–	$0.22268 \dots \geq 2g_{\pi,F}$
$\widehat{\pi}_F$	(6), $= \bar{\pi}_F$	$0.05088 \dots$
$\widehat{g}_F$	(6), $= g_{\pi,F}$	$0.11131 \dots$
$\widetilde{g}_F$	(10), $= g_{\pi,F}$	$0.11131 \dots$
$\text{gain}_\delta(\widetilde{\sigma}_F)$	–	$0.23406 \dots \geq 2g_{\pi,F}$
$\Phi_F(\pi)$	(8), $= 0.4408/g_{\pi,F}$	$3.9600 \dots$
s_F	(11): $\lceil 6.628 \dots \rceil_+ + \lceil 7.187 \dots \rceil - 1$	$7 + 8 - 1 = 14$
$h_{0,F}$	(12): $\lfloor 13.310 \dots \rfloor + 2\lceil 7.187 \dots \rceil$	$13 + 16 = 29$
$h_{1,F}$	(13): $1 + \min\{6.1233 \dots, 4.9600 \dots\}$	$5.9600 \dots$
$b_F^{\max}$	(14): $\lceil 0.929 \dots \rceil - 1$	0
$s_{0,F}$	$s_F + b_F^{\max} h_{0,F}$	14
$s_{1,F}$	$s_F + 1.6/g_{\pi,F}$	$28.3740 \dots$
$s_{2,F}$	(16): $(1 - 0.0622/g_{\pi,F}) + 1.6/g_{\pi,F}$	$0.4412 \dots + 14.3740 \dots$

Proof. The verified inequalities imply (9). Apply theorem 1, whose hypothesis $\sigma < \alpha_\pi$ reads $0.1184 < 0.2$. Because (44) is among what is verified above, $b_F^{\max} = 0$, so $s_{0,F} = s_F = 14$ and $s_{2,F}$ is finite. Substitution of $h_{1,F}$, $s_{1,F}$, $s_{2,F}$ and $h_{0,F} = 29$ in (17) gives (45); also $\alpha_{\pi,F} - \sigma_F = 0.0816$. $\square$

The joint entry of (45) exceeds 1 exactly when $\ell > s_{2,F} + h_{1,F} = 3 + 1.9786/g_{\pi,F}$, and the bracket (43) places that quantity in $(20.761, 20.778)$; the plain ledger entry needs $\ell > s_{1,F} + h_{1,F} = 16 + 2.0408/g_{\pi,F} \in (34.319, 34.337)$ and the constant-charge entry $\ell \geq 43$. All entries are nondecreasing in ℓ , so

$$z_{\min,F}(\ell) > 1 \iff \ell \geq 21 \quad (\ell > 14). \quad (46)$$

The smallest layer count for which the certified latency exceeds $0.2n$ is therefore $\ell = 21$, where the bound is $0.2816n$. The same bracket gives $z_{\min,F}(1000) = 166$, a bound of $13.664n$.

B The Lean development

The development is a single Lean 4 library; running `lake build` in the repository’s `lean/` directory rechecks it. It uses no `sorry` and declares no axioms of its own. Table 2 names, for every numbered item of the paper, the Lean declaration that states or proves it.

Table 2: Every numbered item of the paper and the Lean declaration that states or proves it. Module names are omitted; each declaration is unique in the library.

Paper	Lean
<i>Section 1: model and main result</i>	
(1), expansion profile	LayeredGraph.expands
(2), depth robustness	LayeredGraph.DepthRobust; form NodeDepthRobustAt, deletion depthRobustAt_iff_nodeDepthRobustAt equivalence
(3), pebble budgets	Budget
(4), challenge set	ChallengeBound
(6), $\hat{\pi}$ and $\hat{g}$	Tracking.lam, Tracking.ghat
(9), scalar conditions	GeneralRegime
(10), $\tilde{g}$	Setting.gtilde
(11), search capacity s	sCap, sCapOf, infertileCap, blockedCap
(12), attempt span h_0	h ₀ , growthCap, spendCap
(13), global span h_1	h ₁ , asymptoticGrowth
(14), break capacity	bMax
(15), s_0, s_1	s ₀ , s ₁ , ledgerSlack
(17), chain length	zMin; at $b^{\max} = 0$, zMinNoBreak
(18), path length	latencyLength, latencyLength_general_lower_bound
Theorem 1	latency_general
<i>Section 2: footprint growth and search</i>	
(19), footprint bound	IsFootprintBound, footprintBound; dominance Pebbling.footprintBound_le
(21), expandable level	Expandable
Lemma 1	zetaFloor_le (floor), gtilde_le_gainD_of_infertile (rate), infertile_card_le_gen (capacity)
Lemma 2	infertile_budget
Lemma 4	search_bound, search_reaches, basecase_gen; pebbling half challenge_fertile
Lemma 5	mirror_floor
Lemma 6	growth_window
Lemma 8	fertile_continuation_gen
Lemma 9	break_charge
Lemma 10	extension_attempt_gen
Lemma 3	ChallengeBound.infertile_card_charge
(8), the mid-point $\tilde{\sigma}$	Tracking.mid, mid_gain, two_ghat_le_gainD_of_le_mid
(8), Φ , and the growth term of (13)	growthPot, growthConst
Lemma 7	growthPot_window; (28) is growthConst_window
(27)	growthPot_pi_add_gap; the criterion is growthPot_pi_succ_lt_asymptoticGrowth
<i>Section 3: the latency theorem</i>	

Paper	Lean
Lemma 11	<code>general_ledger</code> (fused; see above)
Lemma 12	the $nb = 0$ clause of <code>GenLedger</code> , proved in <code>general_ledger</code> ; read off by <code>exists_many_links_gen</code>
(16), s_2	s_2 , whose two summands are <code>searchHead</code> (restartHead at a general depth) and <code>jointSlack</code> ; $\min\{\widehat{g}, \widetilde{g}\}$ is <code>gmin</code>
third entry of (17)	<code>jointEntry</code> , whose if on $b^{\max} = 0$ is the $s_2 = \infty$ case; <code>jointRatio_le_zMin</code>
Lemma 13	<code>exists_many_links_gen</code>
Lemma 14	<code>Link.base</code> , <code>Link.extend</code> , <code>chainSystem</code> , <code>chainPathLength</code>
<i>Appendix A: Filecoin parameters</i>	
(42), table 1, the constants	<code>FilecoinLatencyParameters</code> ; <code>sCap_eq</code> , <code>h0_eq</code> , <code>growthConst_eq</code> , <code>h1_eq</code> , <code>bMax_eq_zero</code> , <code>s0_eq</code> , <code>gmin_eq</code> , <code>searchHead_eq</code> , <code>jointSlack_eq</code>
(43), g_π bracket	<code>gpi8_bounds</code>
(44), no-break conditions	<code>chung8GeneralRegime</code>
Corollary 1	<code>latency_corollary</code> ; on the constructed degree-eight curve <code>chung8_latency_corollary</code>
(45), $z_{\min, F}$	<code>filecoinZMin</code> , <code>zMin_eq</code>
(46), onset at $\ell = 21$	<code>filecoinZMin_20_eq</code> , <code>filecoinZMin_21_eq</code> , <code>one_lt_filecoinZMin_iff</code> ; <code>latency_21</code> ; on the con- structed curve <code>chung8_latency_21</code>